



Lessons

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Solving by Factoring

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Solving by Factoring

Toolbox

Solving by Factoring

When you factor an expression, you rewrite it as a product.

$ab + ac \leftarrow$ This expression is the sum of two terms.

$a(b + c) \leftarrow$ The factored expression is the product of a and the quantity $(b + c)$.

Factoring makes it possible to solve quadratic equations because of the **zero property of multiplication**. It states that when the product of two numbers is 0, at least one of the numbers must be 0. For example, if $ab = 0$, then either $a = 0$, $b = 0$, or both equal 0.

Example 1

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Solve for x .

$$x^2 - 2x = 15$$

Step 1

Set the quadratic expression equal to 0 by subtracting 15 from both sides.

$$x^2 - 2x = 15$$

$$x^2 - 2x - 15 = 0$$

Step 2

Factor the left side of the equation.

$$x^2 - 2x - 15 = 0$$

$$(x + 3)(x - 5) = 0$$

Step 3

The quadratic equation has two solutions: $\{-3, 5\}$. Either will make the original equation true.

$$x^2 - 2x = 15$$

$$(-3)^2 - 2(-3) = 15$$

$$9 + 6 = 15$$

$$15 = 15 \quad \text{True}$$

$$x^2 - 2x = 15$$

$$5^2 - 2(5) = 15$$

$$25 - 10 = 15$$

$$15 = 15 \quad \text{True}$$

The two factors are $(x + 3)$ and $(x - 5)$. The equation is true if either factor equals 0. Set the factors equal to 0, and solve for x .

$$x + 3 = 0$$

$$x = -3$$

$$x - 5 = 0$$

$$x = 5$$

Some equations have only one solution.

Example 2

Hide

Solve for m .

$$m^2 - 8m + 16 = 0$$

The equation is already set equal to 0.

Factor the left side:

$$m^2 - 8m + 16 = 0$$

$$(m - 4)(m - 4) = 0$$

Both factors are the same, so there is only one solution.

$$m - 4 = 0$$

$$m = 4$$

The solution set is $\{4\}$.

$$\text{Check: } m^2 - 8m + 16 = 0$$

$$4^2 - 8(4) + 16 = 0$$

$$16 - 32 + 16 = 0$$

$$0 = 0 \quad \text{True}$$

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Category

Lesson 3: Quadratic Equations

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